LAURA CONNOLLY

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CURRENT POSITION

Malone / NSERC Postdoctoral Fellow *Sept. 2025 - present*
Malone Center for Engineering in Healthcare, Johns Hopkins University, Baltimore, USA

- Supervised by Professors Russell H. Taylor and Axel Krieger
- Researching autonomous robotic cancer surgery and ultrasound-guided interventions
- Fellowship announcement available here: malone-postdoctoral-fellow-announcement

EDUCATION

Doctor of Philosophy (Ph.D.) *2020 - 2025*
Queen's University, Kingston, Canada
Electrical Engineering

- Advisors: Dr. Gabor Fichtinger (Queen's), Dr. Parvin Mousavi (Queen's), and Dr. Russell H. Taylor (Hopkins)
- Promoted to Ph.D. without completion of Master's degree
- Specialization in Collaborative Biomedical Engineering
- Vector Institute Trainee
- Member of NSERC CREATE Training Program in Medical Informatics

Bachelor's of Applied Science (B.A.Sc.) *2016 - 2020*
Queen's University, Kingston, Canada
Electrical Engineering

- Graduated with First Class Honours
- Biomedical Engineering Specialization

Relevant Coursework:

- **Fall School on Medical Robotics** *2021*
  Georgia Institute of Technology (Georgia Tech), United States (In-person)
- **Summer School on Image Processing** *2020*
  University of Szeged, Hungary (Online)
- **Hamlyn Winter School on Surgical Imaging and Vision** *2020*
  Imperial College London, United Kingdom (Online)

AWARDS AND SCHOLARSHIPS

Scholarships & grants

- Malone Postdoctoral Fellowship – Johns Hopkins Malone Center for Engineering in Healthcare *2025*
  Competitive fellowship supporting postdoctoral researchers developing engineering solutions for clinical impact at Johns Hopkins.
- NSERC Postdoctoral Fellowship (PDF) *2025*
  Canada's premier federal postdoctoral award, granted nationally to top researchers in natural sciences and engineering.
- Vector Research Grant *2023-2025*
  Research funding from the Vector Institute, Canada's national institute for artificial intelligence research.
- Queen's University Tri-Agency Recipient Recognition Award *2022-2023*
  Institutional award recognizing students who hold major federal Tri-Agency (NSERC/CIHR/SSHRC) scholarships.
- Walter C. Sumner Memorial Fellowship and Renewal *2022-2023*
  Prestigious fellowship awarded to doctoral students in electrical engineering, chemistry, and physics at select Canadian universities.
- NSERC Canada Graduate Scholarship – Doctoral (CGS-D) *2022-2025*
  Canada's federal doctoral scholarship, awarded to the highest-ranked applicants nationwide.
- MediCREATE Training Award *2021*
  Award through an NSERC CREATE training program at Queen's University in medical technology development and translation.
- NSERC Canada Graduate Scholarship – Master's (CGS-M) *2021-2022*
  Canadian federal scholarship for master's students in natural sciences and engineering.

- NSERC Undergraduate Student Research Award (USRA) *2019 & 2020*
  Competitive federal award funding full-time undergraduate research experience.

Research awards

- Governor General's Academic Gold Medal *2026*
  One of Canada's highest academic honors, awarded to the top two graduate students with the most outstanding academic records at their university.
- Best Application Award – Hamlyn Surgical Robotics Challenge *2025*
  International surgical robotics competition held at the Hamlyn Symposium on Medical Robotics, Imperial College London.
- Runner-Up Award – Siemens Healthineers Best Paper (For paper [J-5] at IPCAI) *2025*
  Industry-sponsored best paper award at IPCAI, a leading international conference on computer-assisted interventions.
- Runner-Up – IEEE Ph.D. Research Excellence Award – Kingston Section *2023*
  IEEE regional award recognizing outstanding doctoral research across all IEEE fields.
- Best Demo Award – Advances in Simplifying Medical UltraSound (ASMUS) Workshop – MICCAI *2023*
  Awarded at a workshop of MICCAI, the premier international conference in medical image computing.
- Best Student Paper Award – IEEE International Conferences on Autonomous Systems *2021*
  Top student paper across all submissions to the conference.
- First Place (Student's Choice Award) – Electrical and Computer Engineering Capstone Demo Day *2020*
  Top-ranked final-year engineering design project as voted by peers.
- PEO Student Paper Competition Finalist *2019*
  Provincial technical paper competition run by Professional Engineers Ontario.

Presentation awards

- Best Poster Award (Coauthor & Presenter of [A-4]) – iSMIT *2025*
  International Society for Medical Innovation and Technology, an international conference on emerging medical technologies.
- IPCAI ImFusion Long Abstract Audience Choice Award *2025*
  Audience-voted award for the best long abstract presentation at IPCAI.
- IPCAI J&J MedTech Best Presentation Audience Choice Award *2025*
  Audience-voted award for the best conference presentation at IPCAI.
- Best Presentation Award – ImNO x IGT Symposium *2025*
  Imaging Network Ontario, the annual symposium of Ontario's medical imaging research community.
- Best Presentation Award – ImNO Symposium *2024*
  Imaging Network Ontario, the annual symposium of Ontario's medical imaging research community.
- Runner-Up – Queen's 3 Minute Thesis Competition *2021*
  University-wide research communication competition: an entire thesis presented in three minutes to a general audience.
- Best Pitch Award – ImNO Symposium *2021*
  Top-ranked rapid research pitch at Ontario's annual medical imaging research symposium.

Reviewer awards

- Outstanding Reviewer Award – IPCAI (selected out of 147 reviewers) *2024*
  Recognition for exceptional peer-review quality at a leading computer-assisted interventions conference.
- Outstanding Reviewer Award – MICCAI (selected out of 1659 reviewers) *2023*
  Recognition for exceptional peer-review quality at the premier international medical image computing conference.

Travel grants

- IPCAI Student Participation Award *2025*
  Competitive award supporting student attendance and participation at IPCAI.
- NSERC Michael Smith Foreign Study Supplement (MS-FSS) *2023*
  Federal supplement supporting NSERC scholarship holders undertaking research abroad at leading international institutions.
- SPIE Medical Imaging – Travel Bursary *2020 & 2022 – deferred*
  Travel support for presenting at SPIE Medical Imaging, a major international medical imaging conference.
- Mitacs Globalink Research Award *2020 – deferred to 2021*
  National award funding international research collaborations between Canadian and foreign institutions.

RESEARCH VISITS

Visiting Graduate Scholar *Nov 2021 - Apr 2022 / Mar 2023 - Aug 2023*

Laboratory for Computational Sensing and Robotics (LCSR), Johns Hopkins University, Baltimore, USA

- Supervised by Professors Russell H. Taylor and Simon Leonard
- Supported by a Mitacs Globalink Research Award and NSERC MS-FSS

ACADEMIC POSITIONS

Student Researcher *May 2018 - July 2025*

Laboratory for Percutaneous Surgery (Perk Lab), Queen's University
Medical Informatics Lab (Med-i), Queen's University

- Summer 2018: Reprogrammed a 3D printer to scan tissue samples for data collection
- Summer 2019: Developed, trained, and tested a classifier to distinguish cancer tissue using mass spectrometry
- Graduate studies: Research on robot-assisted breast cancer surgery

Med-i CREATE Program Committee Student Representative *Sept 2022 - Nov 2025*

NSERC CREATE Training Program in Medical Informatics

- Student representative on the Med-i CREATE Program Committee, responsible for attending meetings with the other program members to give feedback on the NSERC CREATE trainee program

Equity, Diversity, Inclusion, and Indigenization (EDII) Officer *Nov 2023 - August 2025*

Perk Lab, Queen's University

- Responsible for organizing seminars related to EDII topics and creating the first lab code of conduct

Technical Lab Coordinator *Jan 2021 - Sept 2023*

Perk Lab, Queen's University

- Technical lab coordinator for the Perk Lab, a research group of approximately 30 undergraduates, engineers, post-doctoral fellows, and graduate students - responsible for acquisitions, expense and resource management

Lab Technician *Jan 2020 - May 2020*

BioRobotics Research Laboratory, Queen's University, Kingston

- Responsible for running tests and supervising the 6 DOF Quanser/ CRS Model A465 Open Architecture Robot for graduate students in ELEC 848: Control System Design for Robots and Telerobots

PROFESSIONAL EXPERIENCE

Medical Systems Engineering Consultant *May 2020 - May 2025*

Medical Robotics Research Group

- Technical consultant and developer for a surgical navigation prototype

Chapter Leader – Makers Making Change Kingston *Jan 2021 – Nov 2021*

A Neil Squire Program

- Chapter leader for the Queen's University chapter of Makers Making Change (MMC), a non-for-profit that 3D prints devices for accessibility

Investment Associate *May 2019 - July 2020*

Front Row Ventures

- Front Row Ventures is the first student-run venture capital fund entirely managed by students in Canada - The investment team is responsible for meeting with student founders and allocating $25000 investments

PATENTS

Photoacoustic tumor bed imaging in surgery

Provisional patent filed Oct 2023

- Inventors: Russell H. Taylor, Emad Boctor, **Laura Connolly**, Parvin Mousavi, Gabor Fichtinger
- More details: https://jhu.technologypublisher.com/technology/53410

PUBLICATIONS

Google Scholar: https://scholar.google.ca/citations?user=9E-xfIwAAAAJ&hl=en

Journal Articles

J-1 **L. Connolly**, H. Song, K. Xu, A. Deguet, S. Leonard, G. Fichtinger, P. Mousavi, R.H. Taylor, E. Boctor. No cancer left behind: A testbed and demonstration of concept for photoacoustic tumor bed inspection. The Journal of Computer Assisted Surgery, Taylor & Francis, 2026 (doi.org/10.1080/24699322.2025.2604123).

J-2 O. Radcliffe, **L. Connolly**, A. Jamzad, M. Kaufmann, S. Merchant, J. Engel, R. Walker, S. Varma, G. Fichtinger, J. Rudan, and P. Mousavi. Anomaly Detection using Intraoperative iKnife Data: A Comparative Analysis in Breast Cancer Surgery. Accepted to the International Journal of Computer Assisted Radiology and Surgery (IJCARS), 2025 (doi.org/10.1007/s11548-025-03476-0).

J-3 **L. Connolly**, T. Ungi, A. Munawar, A. Deguet, C. Yeung, R.H. Taylor, P. Mousavi, G. Fichtinger, K. Hashtrudi-Zaad. Touching the Tumor Boundary: A Pilot Study on Ultrasound-Based Virtual Fixtures for Breast-Conserving Surgery. International Journal of Computer Assisted Radiology and Surgery, special issue of International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2025 (doi.org/10.1007/s11548-025-03342-z). (Shortlisted for Best Paper Award at IPCAI 2025).

J-4 M. Farahmand, A. Jamzad, F. Fooladgar, **L. Connolly**, M. Kaufmann, K. Yi Mi Ren, J. Rudan, D. McKay, G. Fichtinger, P. Mousavi. FACT: Foundation Model for Assessing Cancer Tissue Margins with Mass Spectrometry.International Journal of Computer Assisted Radiology and Surgery, special issue of International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2025 (doi.org/10.1007/s11548-025-03355-8)(Runner-Up Award for Siemens Healthineers Best Paper Award - IPCAI 2025).

J-5 M. Sahu, H, Ishida*, **L. Connolly***, H. Fan*, A. Deguet, P. Kazanzides, F.X. Creighton, R.H. Taylor, A. Munawar. ENTRI: Enhanced Navigational Toolkit for Robotic Interventions. IEEE Transactions on Medical Robotics and Bionics (T-MRB), 2024. 6(4), 1405-1408. (doi.org/10.1109/TMRB.2024.3475827). * denotes equal contribution

J-6 **L. Connolly**, A. Kumar, K.K. Mehta, L. Zogbi, P. Kazanzides, P. Mousavi, G. Fichtinger, A. Krieger, J. Tokuda, R.H. Taylor, Simon Leonard, A. Deguet. SlicerROS2: A research and development module for image-guided robotic interventions. IEEE Transactions on Medical Robotics and Bionics (T-MRB), 2024. 6(4), 1334-1344. (doi.org/10.1109/TMRB.2024.3464683).

J-7 **L. Connolly***, F. Fooladgar*, A. Jamzad*, M. Kaufmann, A. Syeda, K. Ren, P. Abolmaesumi, J.F. Rudan, D. McKay, G. Fichtinger, P. Mousavi. ImSpect: Image-driven Self-supervised Learning for Surgical Margin Evaluation with Mass Spectrometry. International Journal of Computer Assisted Radiology and Surgery, special issue of International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2024. 19(5), 1129–1136. (doi.org/10.1007/s11548-024-03106-1) (Shortlisted for Best Paper Award at IPCAI 2024).

J-8 D. Morton, **L. Connolly**, L. Groves, K. Sunderland, T. Ungi, A. Jamzad, M. Kaufmann, K. Ren, J. Rudan, G. Fichtinger, P. Mousavi. Development of a research testbed for intraoperative optical spectroscopy tumor margin assessment. Acta Polytechnica Hungarica - Special Issue on Digital Health Technologies, 2023 (acta-uni.obuda).

J-9 D. Liu, G. Li, S. Wang, Z. Liu, Y. Wang, **L. Connolly**, G. Shen, K. Cleary, I. Iordachita. An MR Conditional Robot for Lumbar Spinal Injection: Development and Preliminary Validation. The International Journal of Medical Robotics and Computer Assisted Surgery, 2023 (doi.org/10.1002/rcs.2618).

J-10 F. Fooladgar*, A. Jamzad*, **L. Connolly***, A. Santilli, M. Kaufmann, K. Ren, P. Abolmaesumi, J.F. Rudan, D. McKay, G. Fichtinger, P. Mousavi. Uncertainty estimation for margin detection in cancer surgery using mass spectrometry. International Journal of Computer Assisted Radiology and Surgery, 2022. 17(12):2305-13. (doi.org/10.1007/s11548-022-02764-3) * denotes equal contribution

J-11 **L. Connolly**, A. Deguet, S. Leonard, J. Tokuda, T. Ungi, A. Krieger, P. Kazanzides, P. Mousavi, G. Fichtinger, R.H. Taylor. Bridging 3D Slicer and ROS2 for image-guided robotic interventions. J. Sensors, 2022. 14 (5335). (doi.org/10.3390/s22145336) (Highlighted cover-page article)

J-12 **L. Connolly**, A. Jamzad, M. Kaufmann, C.E. Farquharson, K. Ren, J.F. Rudan, G. Fichtinger, P. Mousavi. Combined mass spectrometry and histopathology imaging for perioperative tissue assessment in cancer surgery. J. Imaging, 2021. 7(10), 203. (doi.org/10.3390/jimaging7100203) (Selected out of 27 articles as the cover story for journal issue: www.mdpi.com/2313-433X/7/10)

J-13 A. Santilli, A. Jamzad, N.N.Y Janssen, M. Kaufmann, **L. Connolly**, K. Vanderbeck, A. Wang, D. McKay, J.F. Rudan, G. Fichtinger, P. Mousavi. Perioperative Margin Detection In Basal Cell Carcinoma Using A

Deep Learning Framework: A Feasibility Study. International Journal of Computer Assisted Radiology and Surgery, special issue of International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2020. 15 (5), 887–896. (doi.org/10.1007/s11548-020-02152-9)

Magazine Articles

M-1 O. Radcliffe, A. T. Tun, N. C. Aung, W. L. Thwe, **L. Connolly**, T. Ungi, K. Thornton, C. Davison, E. Purkey, P. Mousavi, and G. Fichtinger. Advancing prosthetic care access on the Thailand–Burma border through open-source technology. IEEE Pulse, 2025.

Full Conference Papers

C-1 K. Hara-Lee, S. Logan, G. Fichtinger, P. Mousavi, A. Deguet, **L. Connolly**. Medical Robot Motion Planner: A Toolkit for Interactive, Patient-Specific Motion Planning in Medical Robotic Interventions. IEEE Engineering in Medicine and Biology Society Conference (EMBC), 2026. (Accepted)

C-2 S. Logan, **L. Connolly**, C. Barr, P. Mousavi, G. Fichtinger and K. Hashtrudi-Zaad. Evaluating Vibrotactile Feedback for Electromagnetic Surgical Navigation. Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 13408. SPIE, 2026. Vancouver, CA. (doi.org/10.1117/12.3087553)

C-3 **L. Connolly**, A. Deguet, A. Munawar, T. Ungi, C. Yeung, R.H. Taylor, P. Mousavi, G. Fichtinger and K. Hashtrudi-Zaad. Fine-Tuning a Virtual Fixture Guidance System for Breast-Conserving Surgery based on Noticeable Differences. Hamlyn Symposium on Medical Robotics (HSMR), 2025. London, UK. (Proceedings: HSMR25-Proceedings-Final.pdf)

C-4 K. Hashtrudi-Zaad, C. Farvolden, **L. Connolly**, C. Barr, and G. Fichtinger. Robotic tracking of a resection cavity using a low cost bench-top robotic arm and electromagnetics. Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 13408. SPIE, 2025. San Diego, USA. (doi.org/10.1117/12.3047143)

C-5 C. Farvolden, K. Hashtrudi-Zaad, **L. Connolly**, C. Barr, G. Fichtinger. An accessible 6-axis testbed for image-guided robotics research. Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 13408. SPIE, 2025. San Diego, USA. (doi.org/10.1117/12.3047507) (3rd Place JMI Pitch Competition)

C-6 D. Liu, Z. Liu, G. Li, **L. Connolly**, G. Shen, and I. Iordachita. Development of an MRI Guided Auxiliary Robot for Spinal Injections. 2023 Seventh IEEE International Conference on Robotic Computing (IRC) (doi.org/10.1109/IRC59093.2023.00066).

C-7 A. Jamzad, F. Fooladgar, **L. Connolly**, D. Srikanthan, A. Syeda, K. Ren, S. Merchant, J. Engel, S. Varma, J.F. Rudan, G. Fichtinger, P. Mousavi. Bridging Ex-Vivo Training and Intra-operative Deployment for Surgical Margin Assessment with Evidential Graph Transformer. 2023 International Conference on Medical Image Computing and Computer-Assisted Intervention (pp. 562-571). Cham: Springer Nature Switzerland (doi.org/10.1007/978-3-031-43990-253)

C-8 **L. Connolly**, A. Deguet, T. Ungi, A. Kumar, A. Lasso, K. Sunderland, P. Kazanzides, A. Krieger, J. Tokuda, S. Leonard, G. Fichtinger, P. Mousavi, R.H. Taylor. An application of SlicerROS2: Haptic latency evaluation for virtual fixture guidance in breast conserving surgery. Hamlyn Symposium on Medical Robotics (HSMR), 2023. London, UK (Proceedings: HSMR23-Proceedings-Final.pdf)

C-9 O. Radcliffe, **Laura Connolly**, T. Ungi, C. Yeo, J.F. Rudan, G. Fichtinger, P. Mousavi. Navigated surgical resection cavity inspection for breast conserving surgery. Medical Imaging 2023: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 12466. SPIE, 2023. San Diego, USA. (doi.org/10.1117/12.2654015)

C-10 D. Morton, **L. Connolly**, L. Groves, K. Sunderland, A. Jamzad, J.F. Rudan, G. Fichtinger, T. Ungi, P. Mousavi. Tracked tissue sensing for tumor bed inspection. Medical Imaging 2023: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 12466. SPIE, 2023. San Diego, USA. (doi.org/10.1117/12.2654217)

C-11 **L. Connolly**, A. Jamzad, A. Nikniazi, R. Poushimin, J.M. Nunzi, J.F. Rudan, G. Fichtinger, P. Mousavi. Feasibility of combined optical and acoustic imaging for surgical cavity scanning." Medical Imaging 2022: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 12034. SPIE, 2022. San Diego, USA. (doi.org/10.1117/12.2611964)

C-12 A. Greisman, A. Ivimey, **L. Connolly**, K. Hashtrudi-Zaad. Power based gravity compensation for flexible joint manipulators. IEEE International Symposium on Measurement and Control in Robotics (ISMCR), 2022. (doi.org/10.1109/ISMCR56534.2022.9950579)

C-13 **L. Connolly**, A. Deguet, K. Sunderland, A. Lasso, T. Ungi, J.F. Rudan, R.H. Taylor, P. Mousavi, G. Fichtinger. An Open-Source Platform for Cooperative Semi-Autonomous Robotic Surgery. 2021 IEEE International Conference on Autonomous Systems (ICAS), pp. 1–5, Aug. 2021. (doi.org/10.1109/ICAS49788.2021.9551149) (Best Student Paper Award)

C-14 F. Akbarifar, A. Jamzad, A. Santilli, M. Kaufmann, N. Janssen, **L. Connolly**, K. Ren, K. Vanderbeck, A. Wang, D. McKay, J.F. Rudan, G. Fichtinger, P. Mousavi. Graph-based analysis of mass spectrometry data for tissue characterization with application in basal cell carcinoma surgery. Medical Imaging 2021: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 11598. SPIE, 2021. (doi.org/10.1117/12.2582045)

C-15 **L. Connolly**, A. Jamzad, M. Kaufmann, R. Rubino, A. Sedghi, T. Ungi, M. Asselin, S. Yam, J.F. Rudan, C. Nicol, G. Fichtinger, P. Mousavi. Classification of tumor signatures from electrosurgical vapors using mass spectrometry and machine learning: a feasibility study." Medical Imaging 2020: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 11315. SPIE, 2020. Houston, USA. (doi.org/10.1117/12.2549343)

C-16 L. Yates, **L. Connolly**, A. Jamzad, M. Asselin, R. Rubino, S. Yam, T. Ungi, A. Lasso, C. Nicol, P. Mousavi, G. Fichtinger. Robotic tissue scanning with biophotonic probe. Medical Imaging 2020: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 11315. SPIE, 2020. Houston, USA. (doi.org/10.1117/12.2549635)

C-17 **L. Connolly**, T. Ungi, A. Lasso, T. Vaughan, M. Asselin, P. Mousavi, S. Yam, G. Fichtinger. Mechanically controlled spectroscopic imaging for tissue classification. Medical Imaging 2019: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 10951. SPIE, 2019. San Diego, USA. (doi.org/10.1117/12.2512481)

Conference Abstracts

A-1 **L. Connolly**, J. Ge, R.H. Taylor, A. Krieger. Integrating ultrasound into autonomous robotic workflows for cancer surgery. The 9th Annual Johns Hopkins Research Symposium on Engineering in Healthcare, 2025.

A-2 **L. Connolly**, G. d'Albenzio, R. Hisey, T. Ungi, F. Saleck, P. Mousavi, R. Kikinis, Y. Ould Tfeil, A. Moulaye Idriss, G. Fichtinger. Print-and-Poke: A Low-Cost 3D-Printed Guidance Solution for Percutaneous Nephrostomy in LMICs. Hamlyn Surgical Robotics Challenge 2025. (Best Application Award)

A-3 M. Bernardes, P. Moreira, **L. Connolly**, A. Deguet, S. Leonard, J. Tokuda. Integration of a ROS2-Based Robotic System with MRI-Guided Percutaneous Intervention Workflow. 36th Annual International Society for Medical Innovation and Technology Conference (iSMIT 2025), 2025, Washington, DC, USA.

A-4 A. Jamzad, M. Kaufmann, **L. Connolly**, T. Ungi, C. Yeung, T. Koster, K. Yi Mi Ren, S. Merchant, C. Jay Engel, G. Ross Walker, A. Gallo, D. Jabs, S. Varma, J. F. Rudan, G. Fichtinger, P. Mousavi. Visualizing Cancer Margins in Breast Conserving Surgery Using Intraoperative Mass Spectrometry. 36th Annual International Society for Medical Innovation and Technology Conference (iSMIT 2025), 2025, Washington, DC, USA. (Best Poster Award)

A-5 J. Tokuda, P. Moreira, M. Bernardes, **L. Connolly**, A. Kumar, L. Al-Zogbi, L. Foley, K. Tuncali, C. Tempany, N. Hata, I. Iordachita, A. Krieger, A. Deguet, S. Leonard. Open-Source 3D-Printable MRI-Conditional Needle Placement Robot for Prostate Interventions. International Society for Magnetic Resonance in Medicine, Annual Meeting, 2025.

A-6 E. Elkind, O. Radcliffe, A. Tin Tun, **L. Connolly**, C. Davison, E. Purkey, G. Fichtinger, K. Thornton. Strengthening Low-cost Prosthetic Solutions in Thailand/Myanmar Through Academic Institution-NGO Collaboration. Health and Human Rights Conference, Queen's University, 2025. (3rd Place Research Competition)

A-7 **L. Connolly**, T. Ungi, A. Munawar, A. Deguet, C. Yeung, R.H. Taylor, P. Mousavi, G. Fichtinger, K. Hashtrudi-Zaad. A feasibility study on enhanced navigation in breast-conserving surgery through haptic feedback. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium. (Best Presentation Award)

A-8 K. Hashtrudi-Zaad, C. Farvolden, **L. Connolly**, C. Barr, G. Fichtinger. Resection cavity tracking using a bench-top robot and electromagnetic tracking. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-9 C. Farvolden, K. Hashtrudi-Zaad, **L. Connolly**, C. Barr, G. Fichtinger. Designing a 6-Axis Testbed for Accessible Image-Guided Robotics Research. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-10 T. Elliott, F. Fooladgar, A. Jamzad, **Laura Connolly**, M. Kaufmann, K. Ren, J. Rudan, D. McKay, G. Fichtinger, and P. Mousavi. A Comparison of Uncertainty Techniques on Basal Cell Carcinoma Mass Spectrometry Data. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-11 E. Elkind, A. Tin Tun, O. Radcliffe, **L. Connolly** , C. Davison, E. Purkey, P. Mousavi, G. Fichtinger, K. Thornton. Developing low-cost 3D-printed prosthetics with a functional wrist for patients along the Thai-Myanmar border. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-12 M. Farahmand, A. Jamzad, F. Fooladgar, **L. Connolly**, M. Kaufmann, K. Yi Mi Ren, J.F. Rudan, D. McKay, G. Fichtinger, P. Mousavi. Foundation Models for Cancer Tissue Margin Assessment with Mass Spectrometry. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-13 E. Elkind, A. Tin Tun, O. Radcliffe, **L. Connolly**, C. Davison, E. Purkey, P. Mousavi, G. Fichtinger, and K. Thornton. Enhancing healthcare access by developing low-cost 3D printed prosthetics along the Thai-Myanmar border. The Canadian Conference on Global Health, 2024.

A-14 **L. Connolly**, A. Jamzad, M. Kaufmann, D. McKay, G. Fichtinger, P. Mousavi. Real-time margins for skin cancer excision using mass spectrometry analysis of cautery smoke. The 77th Annual Meeting of the Canadian Society of Plastic Surgeons, 2024.

A-15 **L. Connolly**, H. Song, K. Xu, A. Deguet, S. Leonard, G. Fichtinger, P. Mousavi, R.H. Taylor, E. Boctor. Photoacoustic detection of residual cancer in breast-conserving surgery. The 22nd Annual Symposium of the Imaging Network of Ontario (ImNO), 2024. (Best Presentation Award)

A-16 O. Radcliffe, A. Tin Tun, N. Chan, **L. Connolly**, T. Ungi, K. Thornton, C. Davison, E. Purkey, P. Mousavi, G. Fichtinger. 3D printing prosthetics on the Thailand-Myanmar border. The 22nd Annual Symposium of the Imaging Network of Ontario (ImNO), 2024.

A-17 **L. Connolly**, A. Deguet, T. Ungi, A. Kumar, A. Lasso, K. Sunderland, P. Kazanzides, A. Krieger, J. Tokuda, S. Leonard, G. Fichtinger, P. Mousavi, R.H. Taylor. Haptic feedback and ultrasound guidance for lumpectomy with SlicerROS2. ASMUS workshop at MICCAI, 2023. (Best Demo Award)

A-18 **L. Connolly**, A. Jamzad, F. Fooladgar, A. Syeda, M. Kaufmann, K. Ren, P. Abolmaesumi, J.F. Rudan, D. McKay, G. Fichtinger, P. Mousavi. Margin detection in skin cancer surgery via 2D representation of mass spectrometry data. The 21st Annual Symposium of the Imaging Network of Ontario (ImNO), 2023.

A-19 D. Morton, **L. Connolly**, L. Groves, K. Sunderland, A. Jamzad, J.F. Rudan, G. Fichtinger, T. Ungi, P. Mousavi. Evaluation of tracked optical tissue sensing for tumor bed inspection. The 21st Annual Symposium of the Imaging Network of Ontario (ImNO), 2023.

A-20 O. Radcliffe, **L. Connolly**, T. Ungi, C. Yeo, J.F. Rudan, G. Fichtinger, P. Mousavi. Electromagnetic navigation for residual tumor localization in breast-conserving surgery. The 21st Annual Symposium of the Imaging Network of Ontario (ImNO), 2023.

A-21 A. Jamzad, **L. Connolly**, F. Fooladgar, M. Kaufmann, K. Ren, S. Merchant, J. Engel, S. Varma, P. Abolmaesumi, G. Fichtinger, J.F. Rudan, P. Mousavi. Deployment of deep models for intra-operative margin assessment using mass spectrometry. Neural Information Processing Systems (MED-NEURIPS), 2022.

A-22 **L. Connolly**, A. Jamzad, A. Nikniazi, R. Poushimin, A. Lasso, K. Sunderland, T. Ungi, J. Michel Nunzi, J.F. Rudan, G. Fichtinger, P. Mousavi. An open-source testbed for developing image-guided robotic tumor-bed inspection. The 20th Annual Symposium of the Imaging Network of Ontario (ImNO), 2022.

A-23 **L. Connolly**, K. Sunderland, A. Deguet, A. Lasso, T. Ungi, J.F. Rudan, R.H. Taylor, P. Mousavi, G. Fichtinger. A platform for robot-assisted intraoperative imaging in breast conserving surgery. The 19th Annual Symposium of the Imaging Network of Ontario (ImNO), 2021. (Best Pitch Award)

A-24 **L. Connolly**, A. Jamzad, M. Kaufmann, R. Rubino, A. Sedghi, T. Ungi, M. Asselin, S. Yam, J.F. Rudan, C. Nicol, G. Fichtinger, P. Mousavi. Classification of primary tumor and surrounding tissue in breast cancer xenograft models. The 18th Annual Symposium of the Imaging Network of Ontario (ImNO), 2020.

TEACHING EXPERIENCE

Teaching Assistantships

Queen's University

- Graduate TA - Artificial Intelligence *Jan 2025 - March 2025*
- Graduate TA - Introduction to Robotics *Jan 2024 - May 2024*
- Graduate TA - Electrical and Computer Engineering Design and Practice *Sept 2023 - Dec 2023*
- Graduate TA - Electrical and Computer Engineering Design Project (Capstone) *Sept 2020 - May 2023*
- Project Manager - Engineering Design and Practice Sequence (Module 3) *Sept 2019 – Apr 2020*

Healing Through Collaboration: Open-Source Software in Surgical, Biomedical, and AI Technologies
Hamlyn Symposium for Medical Robotics *2025-2026*

- As a co-organizer of this workshop at HSMR 2025 & 2026, I contributed to speaker coordination, poster review, general planning, and building the website. More details are available here: healing-through-collaboration

Workshops on Open-Source Software for Image-Guided Robotics
International Symposium for Medical Robotics *2023-2026*

- As a member of the ROS-MED research team, I have helped organize and present four workshops to teach members of the community how to build open-source image-guided robotic systems (event-history)
- In our most recent session at ISMR 2026, I led a tutorial on motion planning in SlicerROS2, available online here: tutorial-recording
- We have now extended our workshop offerings and will be hosting an interactive session at IEEE EMBC 2026: advertisement

Electrifying Medicine
Queen's University *April 2024*

- Organized a bilingual, hands-on session for 30 local grade six students on robotics and the basics of electricity

MD Meets Machine
Queen's University *July 2022 - Aug 2022*

- Helped run and prepare a workshop for high-school students interested in medical image computing research
- More details on this event are available here: news-article

MENTORSHIP EXPERIENCE

Below is a list of projects on which I have served as a mentor in an official capacity:

1. **Autonomous robotic partial nephrectomy** (Feb. 2026 - *Apr. 2026*)
   I co-mentored three student groups for the Advanced Computer-Integrated Surgery (CIS II) course at Johns Hopkins University. There were five students in total working on various projects towards achieving autonomous partial nephrectomy. The topics under investigation included simulation, bleeding control, and anatomy tracking in a minimally invasive surgical setup.
2. **Wearable haptic feedback for tumor localization** (May 2025 – *present*)
   I secured funding to support another undergraduate mechatronics student developing a wearable haptic feedback bracelet for tumor localization in breast surgery. Their work was accepted for presentation at SPIE Medical Imaging 2026 (C-2).
3. **Medical robot motion planning** (Sept. 2024 – *present*)
   I am currently mentoring an M.A.Sc. student in the Perk / Med-i Lab who is optimizing a motion planning system for image-guided robotic applications. Their work was just accepted to the IEEE EMBC Conference for 2026 (C-1).
4. **Autonomous ultrasound scanning** (Sept. 2024 – March 2025)
   I mentored two projects (Grade 5 & 7) for the Frontenac, Lennox, and Addington Science Fair. Both projects won special awards and one won a silver medal in the engineering category.
5. **Robotic tumor bed inspection** (May 2024 – Aug. 2024)
   I was awarded a Med-i CREATE Summer Research Award to support an undergraduate mechatronics student in summer 2024. We were also joined by a student from the Engineering Science program at the University of Toronto. Together, we developed a low-cost bench-top system for tumor bed inspection using a 6-axis robot. Each student had a first-author paper accepted and presented at the SPIE Medical Imaging Conference (C-4, C-5), and one student won third place in the JMI pitch competition.
6. **3D-printed prosthetics for Burma Children's Medical Fund** (May 2023 – *present*)
   I helped initiate a collaboration with the BCMF clinic in Thailand, which serves refugees of the Burma civil war by providing 3D-printed prosthetics to patients lacking conventional healthcare access. We have since sent two students to the clinic, whom I trained in 3D printing and prototyping and supported throughout their time abroad. I have also contributed to several abstract submissions related to this work (A-6, A-11, A-13, A-16). This work received third place at the Health and Human Rights Conference at Queen's University, two prizes totaling $4,000 USD at the Rice360 Global Health Technologies Design Competition, and first place overall ($1000 USD) at the RESNA conference in Chicago.

7. **Tracked ultrasound imaging for tumor bed inspection** (May 2022 - Aug. 2022)
I mentored an undergraduate Biomedical Computing student through the Google ExploreCSR program (news-article). We developed a prototype navigation system for ultrasound-guided tumor bed inspection, which they presented as an oral presentation at the SPIE Medical Imaging Conference in 2023 (C-8). They went on to complete their Master's, where I continued to mentor them and we published a journal article together (J-3). They are now working at ClaroNav.

8. **Spectroscopic tissue scanning** (Sept. 2021 – Sept. 2023)
I mentored an M.A.Sc. student who continued my work (presented in C-10) on tissue classification using spectroscopy. Together, we explored techniques for tracked scanning of ex vivo tissue and machine learning-based spectral classification. This collaboration resulted in both a conference (C-9) and a journal publication (J-9). They are now working at Medtronic.

SELECTED WORKSHOPS AND INVITED TALKS

Please note that some of the presentation titles are hyperlinked to articles about or recordings of the presentation.

Guest Lecturer - Biomedical Robotics — ***March 2026***
Worcester Polytechnic Institute (Virtual)

Guest Lecturer - CIS II: Advanced Computer-Integrated Surgery — ***February 2026***
Johns Hopkins University

Guest Lecturer - CIS I: Computer-Integrated Surgery — ***October 2025***
Johns Hopkins University

Virtual Open Forum on Computer- and Robotics-Assisted Cancer Surgery — ***September 2025***
Machine Learning in Medical Imaging Consortium (MaLMIC)

Rising Star Talk — ***May 2025***
ICRA Robot-Assisted Medical Imaging Workshop (RAMI)

Keynote Speaker - STEM Horizons — ***March 2025***
AI & Healthcare Summit

Debate Participant: The Role of AI in Medicine — ***March 2024***
Imaging Network of Ontario Symposium / NSERC CREATE Program for Medical Informatics and Responsible AI

Guest Lecturer - CISC 330: Computer Integrated Surgery — ***Nov 23 2023***
Queen's University, School of Computing

Invited Speaker: Open-source software for surgical technologies workshop — ***June 2023***
Hamlyn Symposium on Medical Robotics (HSMR), London, UK

Presenter: Building Software Systems for Image-Guided Robot-Assisted Interventions — ***April 2023***
International Symposium on Medical Robotics (ISMR), Atlanta, USA

Invited Panelist — ***March 2021***
Queen's Women in Engineering and Applied Science Conference (Q-WASE)

PROFESSIONAL SERVICE

Reviewer for: Nature, IEEE Transactions on Biomedical Engineering, IEEE Robotics and Automation Letters, IEEE International Conference on Intelligent Robots and Systems (IROS), SPIE Journal of Electronic Imaging, SPIE Medical Imaging Conference, Medical Image Computing and Computer Aided Intervention Conference (MICCAI), International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), The Hamlyn Symposium on Medical Robotics (HSMR), IEEE International Conference on Robotics and Automation (ICRA)

Scholarship Committees: Vector Scholarship in AI, NSERC Scholarships and Fellowships Selection Committee for Computing Sciences (198)

Area Chair: IPCAI 2026